A MAXIMAL NON-SCHEME FIBRATION IDEAL EXISTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4124-4137, following the statement at 4114-4122.

Editorial note. No omissions.

Allowed standard prerequisites (editorial).

- Limits that are schemes. lemma-limit-is-scheme (same chapter, cited at ~4131-4132). If X_{A/I} were a scheme, one of the X_{A/I_alpha} would be a scheme.

Source reference. Original lemma 07SR: sources/stacks-spaces-limits/pinned-original-source.tex, source lines 1435–1435. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\varphi : X \to \mathrm{Spec}(A)$ be a quasi-compact and quasi-separated morphism from an algebraic space to an affine scheme. If X is not a scheme, then there exists an ideal $I \subset A$ such that the base change $X_{A/I}$ is not a scheme, but for every $I \subset I'$, $I \neq I'$ the base change $X_{A/I'}$ is a scheme.*

Proof. We prove this by Zorn's lemma. Let $\mathcal{I}$ be the set of ideals I such that $X_{A/I}$ is not a scheme. By assumption $\mathcal{I}$ contains (0). If I_α is a chain of ideals in $\mathcal{I}$, then $I = \bigcup I_\alpha$ is in $\mathcal{I}$. Namely, $A/I = \mathrm{colim}\, A/I_\alpha$, hence

$$X_{A/I} = \lim X_{A/I_\alpha}$$

Thus we may apply Lemma 07SR to see that if $X_{A/I}$ were a scheme, then so would be one of the X_{A/I_α}. Thus $\mathcal{I}$ has maximal elements by Zorn's lemma. □

Date: stacks-project:spaces-limits.tex:local-snapshot-20261012; GNU FDL 1.2 or later.